

Cost estimate

Token volumes below are measured against the real corpus and real production chat turns — not estimates. Dollar figures are left as `<rate>` placeholders: this is a training-program Foundry resource, not a standard pay-as-you-go subscription, so list pricing from the public Azure calculator may not reflect the actual billing rate. Plug in the current per-1M-token rate from your Azure pricing page for `text-embedding-3-large`, GPT-5, GPT-5.5, and DeepSeek V3.2 to get dollar figures.

Methodology

- **Ingestion (one-time + per re-index)**: measured by running the real chunker against the current 24-document corpus and counting tokens with the same `cl100k_base` encoding used for chunking: **222 chunks, 23,378 embedding tokens total** (avg 105.3 tokens/chunk). Re-measured 2026-07-22 after 4 documents were added to the original 20-document Contoso set (see `design-doc.md` §2) — up from 179 chunks / 18,639 tokens, proportional to the 4 added documents. Re-index frequency assumed monthly unless the corpus churns faster.
- **Chat (recurring, scales with users)**: measured real end-to-end turns from the live `messages` table in production, against the current corpus, all three selectable models:
- **GPT-5** (9 turns measured): avg input 1,216 tokens, avg output 178 tokens — **avg total 1,394 tokens/turn**.
- **GPT-5.5** (6 turns measured): avg input 1,463 tokens, avg output 151 tokens — **avg total 1,613 tokens/turn**.
- **DeepSeek V3.2** (6 turns measured): avg input 1,360 tokens, avg output 97 tokens — **avg total 1,456 tokens/turn** — the shortest average output of the three, a real measured verbosity difference, not just a cost one.
- Plus one embedding call per question (~10-50 tokens, negligible).
- All three reflect the `reasoning.effort` / `text.verbosity` tuning established 2026-07-20 (`"minimal"` / `"low"` for GPT-5, `"none"` / `"low"` for GPT-5.5; DeepSeek accepts neither param).
- This is the dominant cost driver at scale, not storage or retrieval — context size (top-k × avg chunk size) matters more than anything else in this line item, which is why top-k=5 and structure-aware chunking were chosen deliberately (see pattern-justification table).
- **Retrieval**: Azure AI Search pricing is tier/replica/partition-based, not per-query — cost is a step function of scale (upgrade tier or add replicas), not linear per-user cost.
- **Storage**: production now runs on Azure Database for PostgreSQL Flexible Server (see `design-doc.md` §4); local dev still uses Docker Postgres at no cost. Storage cost for this data volume (conversations/messages/citations, no document bodies) is negligible relative to generation.

Cost per 1,000 chat turns (3-model comparison)

	GPT-5	GPT-5.5	DeepSeek V3.2
Avg tokens/turn	1,394	1,613	1,456
Tokens per 1,000 turns	1.394M	1.613M	1.456M
Cost per 1,000 turns	$1.394 \times \text{<rate>}$	$1.613 \times \text{<rate>}$	$1.456 \times \text{<rate>}$
Foundry RPM quota	150,000	5,000	500
Foundry TPM quota	15M	5M	500K

GPT-5.5 costs ~16% more per turn than GPT-5 on these measured samples; DeepSeek sits in between on tokens/turn but has by far the weakest quota (500 RPM — 300x less than GPT-5), which is why it was rejected in [design-doc.md](#) §5 despite being cost-competitive per turn. Quota, not per-turn token cost, is what actually gates which model can serve 5,000 users.

100 users vs. 5,000 users

Assumes 5 chat turns/user/day (an internal HR assistant, not a high-frequency tool), GPT-5 as the default model.

	100 users	5,000 users
Turns/day	500	25,000
Generation tokens/day	$500 \times 1,394 = 697,000$	$25,000 \times 1,394 = 34,850,000$
Generation cost/day	$0.697 \times \text{<rate per 1M tokens>}$	$34.85 \times \text{<rate per 1M tokens>}$
Peak RPM (worst case: all users, 1 turn/min)	100	5,000
GPT-5 RPM headroom	$100 / 150,000 = 0.07\%$	$5,000 / 150,000 = 3.3\%$
AI Search tier needed	Basic (semantic ranker required)	Basic — same tier scales via replicas, not a tier change, until query volume is much higher
Embedding cost (one-time full re-index)	$0.0234 \times \text{<rate per 1M tokens>}$ (same for both — independent of user count)	

Takeaway: generation is still the dominant recurring cost and still scales linearly with turns; GPT-5's quota headroom is effectively unchanged (3.3% of 150K RPM at 5,000 users). Real measured data across all three selectable models now backs the model-choice reasoning in [design-doc.md](#) §5 on cost, verbosity, and quota simultaneously, not quota alone.